Latest Advancements (2025-2026)

ADVANCEMENT	WHAT HAS CHANGED	SMT APPLICATION/BENEFIT
AI-assisted AXI inspection	AI/deep learning is increasingly used for defect classification, image segmentation, and verification rather than relying only on rule-based algorithms.	Lower false calls, better defect classification and less manual review, especially for complex assemblies.
AI-assisted inspection-program creation	AI can automatically identify components/solder joints and assist in generating or optimising inspection programs.	Major reduction in programming/NPI time; useful for high-mix SMT.
Faster 3D/CT AXI	New acquisition and reconstruction methods are increasing 3D inspection speed without sacrificing image quality.	Makes 3D CT more practical for inline production and higher inspection coverage.
Dynamic Planar CT	Newer planar-CT approaches dynamically acquire X-ray data and use improved 3D reconstruction algorithms.	Faster inspection of dense PCBs while retaining layer separation and 3D information.
Multi-resolution X-ray imaging	Modern systems can switch between resolution levels based on inspection requirements.	High-resolution inspection of fine-pitch areas while maintaining production throughput elsewhere.
Higher-throughput dynamic 3D acquisition	Dynamic image acquisition is becoming a key differentiator, rather than increasing X-ray resolution alone.	Allows 3D AXI to be used for more demanding inline/100% inspection applications.
Large-board/long-board AXI	AXI systems are expanding to handle substantially larger assemblies.	Server boards, telecom/5G, LED, semiconductor and other large-format electronics.
Improved inspection of complex/high-density assemblies	Better 3D reconstruction, dynamic acquisition, and powerful microfocus X-ray sources are improving inspection even in assemblies with severe component shadowing.	BGA, LGA, QFN, BTC, PoP and double-sided/high-density PCB assemblies.
AI-based image segmentation	AI is being applied to difficult image-processing problems such as locating/segmenting voids and other structures even when the X-ray image is distorted.	More reliable void analysis and defect detection in challenging assemblies.
AXI + factory/process data integration	Inspection data is increasingly being combined with data from other SMT processes rather than remaining an isolated inspection result.	Moves from 'detect defect' → 'identify process cause' → 'correct process' and supports closed-loop manufacturing.

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